TOPIC 01 · FUNDAMENTALS OF GOOGLE SHEET
LAB 4 THEME · FORMULAS: VLOOKUP, COUNTIF & NESTED IF

Lab 4 — Generate, Explain and Fix Screening Formulas

Maps to	LO1: use Gemini to generate VLOOKUP, IF, nested IF and COUNTIF/COUNTIFS formulas from plain-English rules, explain them step by step and fix formula errors, verifying the results on one row and in total
Duration	30 minutes
Tools	Google Sheets · Gemini formula help (side panel, = then Cmd/Ctrl + Ctrl/Alt + G, Fix) · OrchidBay_Screening_Formulas.xlsx
Mock data	data/OrchidBay_Screening_Formulas.xlsx
Course	Gemini for Google Sheet (C817) · v2.0
Prompts	https://tertiarycourses.github.io/C817-Gemini-for-Google-Sheet/prompts/#lab-4

Goal

Add four screening columns to the clean applications - Budget_Max, Min_Years, Salary_Fit and Screen_Result - and prove they are right.

What You Will Do

The hiring managers want a first screen of all 240 applications. You will fix a broken lookup with Gemini's Fix button, then describe each rule in plain English and let Gemini write the formula: a VLOOKUP for the role's minimum experience, an IF for salary fit, and a nested IF with AND that labels each candidate Strong, Possible or Not suitable. Gemini explains the formula, COUNTIF and COUNTIFS summarise the results, and independent check formulas prove them.

What You Will Build

OrchidBay_Screening_Formulas with working Budget_Max, Min_Years, Salary_Fit and Screen_Result columns, a Checks tab of COUNTIF/COUNTIFS results that reconcile to 240, and a filled Prompt_Log

Prerequisites

- Labs 1-3 completed, or familiarity with the Applications columns (see About_This_Data).
- This lab folder downloaded, including data/OrchidBay_Screening_Formulas.xlsx.

Data note

Use only the supplied fictional Orchid Bay data or information you are authorised to process. Do not paste credentials, confidential business material or real candidates' or employees' personal data into Gemini, the AI function or a Gem unless your organisation has approved it.

Steps

Step 1. Download OrchidBay_Screening_Formulas.xlsx from this lab's data folder. In Google Drive choose New > File upload, open the uploaded file, then choose File > Save as Google Sheets - Gemini works best on a native Google Sheets file. Close the .xlsx tab and work only in the new Google Sheets copy. At the top right, click Ask Gemini (the spark icon) to open the Gemini side panel. Read the About_This_Data tab before you prompt.

PROMPT / CHECK — copy and adapt

```
Required state:
File : OrchidBay_Screening_Formulas (Google Sheets copy)
Tabs : About_This_Data | Applications | Jobs | Settings | Checks | Prompt_Log
Settings : B1 report date 2026-09-30 | B2 pass mark 70 | B3 possible mark 60
```

Step 2. Column Q (Budget_Max) shows an error. Hover over Q2 and click Fix: Gemini reviews the formula and explains the problem - VLOOKUP asks for column 9 of a range that has only 8 columns. Accept the fix, then copy Q2 down to Q241 (or double-click the fill handle). Q2 should show S$6,500. If you don't see Fix, ask Gemini instead.

PROMPT / CHECK — copy and adapt

```
Before : =VLOOKUP(E2,Jobs!$A$2:$H$9,9,FALSE) -> #REF!
After : =VLOOKUP(E2,Jobs!$A$2:$H$9,8,FALSE) -> S$6,500

ASK GEMINI:
Explain why the formula in Q2 shows #REF! and give me the corrected formula.
```

Step 3. Type Min_Years in R1. Click R2, then ask Gemini for the formula - in the side panel, or type = in the cell and press Cmd + Ctrl + G (Mac) / Ctrl + Alt + G (Windows). Click Insert, then fill it down to row 241. R2 should show 2 (Data Analyst).

PROMPT / CHECK — copy and adapt

```
FORMULA:
Create a formula for R2 that looks up the Job_ID in E2 on the Jobs tab (A2:H9) and returns
Min_Years.
```

Step 4. Type Salary_Fit in S1, ask for the formula in S2, insert it and fill it down. S2 should say Within budget (S$6,200 against a Budget_Max of S$6,500).

PROMPT / CHECK — copy and adapt

```
FORMULA:
Create a formula for S2 that returns "Within budget" if Expected_Salary in K2 is less than or
equal to Budget_Max in Q2, otherwise "Above budget".
```

Step 5. Type Screen_Result in T1. Describe the whole rule in one prompt - Gemini nests the IFs for you. Insert and fill down. T2 (CAN-26481, score 65, 2 years) should say Possible. Screen_Result only flags candidates for a recruiter to review - a person still reads every application before anyone is rejected.

PROMPT / CHECK — copy and adapt

```
FORMULA:
Create a formula for T2: if Screening_Score in M2 is at least the pass mark in Settings!B2 and
Years_Experience in H2 is at least Min_Years in R2, return "Strong"; otherwise, if M2 is at
least Settings!B3, return "Possible"; otherwise return "Not suitable".
```

Step 6. Ask Gemini to explain the formula. Check its explanation against T2's actual values in M2, H2 and R2 - an explanation you can't match to the data is not an explanation.

PROMPT / CHECK — copy and adapt

```
ASK GEMINI:
Explain the formula in T2 step by step, and tell me what it returns for the candidate in row 2
and why.
```

Step 7. On the Checks tab, put labels in column A and ask Gemini for these two formulas one at a time, inserting each into column B. Expect 88 Strong in total and 14 Strong Data Analysts.

PROMPT / CHECK — copy and adapt

```
FORMULA:
1) Create a formula that counts how many cells in Applications!T2:T241 are "Strong".
2) Create a formula that counts the rows on Applications where Position is "Data Analyst" and
Screen_Result is "Strong".
```

Step 8. Paste the independent checks below them. Strong + Possible + Not suitable must add up to 240, and the Above budget count must match a COUNTIF of Salary_Fit. Then ask Gemini a what-if question and test its answer: change Settings!B2 to 75 - the Strong count should become 60 - then set it back to 70.

PROMPT / CHECK — copy and adapt

```
CHECK:
Possible =COUNTIF(Applications!T2:T241,"Possible")
Not suitable =COUNTIF(Applications!T2:T241,"Not suitable")
All results =COUNTA(Applications!T2:T241)
Above budget =SUMPRODUCT(--(Applications!K2:K241>Applications!Q2:Q241))

ASK GEMINI:
If I change the pass mark in Settings!B2 from 70 to 75, how many candidates will be Strong?
```

Step 9. Prompt_Log already lists every prompt in this lab with its step. Ask Gemini to recap the conversation, copy each line into What Gemini did, add How I verified in your own words, and choose Keep, Revise or Undo in the Decision dropdown. Google Sheets saves automatically. Tip: the side-panel history is lost when you reload the page, so insert or copy anything you want to keep.

PROMPT / CHECK — copy and adapt

```
LOG:
List the prompts I sent you in this conversation. For each one, give a one-line summary of
what you did, so I can copy it into the Prompt_Log tab.
```

Sample Prompts

Use these prompts in the Gemini side panel in Google Sheets (or adapt them). Name the tab and columns, state the output, and always verify the result. The same prompts are in sample-prompts.txt in this lab folder.

Copy-ready prompts online: https://tertiarycourses.github.io/C817-Gemini-for-Google-Sheet/prompts/#lab-4

#	Purpose	Prompt
1	Lookup	Create a formula that returns the Hiring_Manager for the Job_ID in E2 from the Jobs tab (A2:J9).
2	Date	Create a formula for the number of days between Applied_Date in D2 and the report date in Settings!B1.
3	Text	Create a formula that returns TRUE if Key_Skills in J2 contains "SQL".
4	Count	Create a formula that counts Above budget candidates for each Position.
5	Explain	Explain the difference between VLOOKUP and XLOOKUP, and when to use each.

Test It

Check	Expected result
Fix	Q2 = S$6,500; no #REF! left in column Q.
Row 2	CAN-26481: Min_Years 2, Salary_Fit Within budget, Screen_Result Possible.
Screen_Result totals	Strong 88 + Possible 73 + Not suitable 79 = 240.
COUNTIFS	Strong Data Analysts = 14; Above budget = 62 (Within budget 178).
What-if	Pass mark 75 gives 60 Strong; set back to 70 gives 88.

Checkpoint

Topic 1 ends here. Lab 5 starts Topic 2 with its own tracker workbook.

Troubleshooting

If this happens	Fix
The Fix button does not appear.	Click the cell and hover over the red triangle. If there is still no Fix, use the ASK GEMINI prompt in Step 2 and type the corrected formula yourself.
Gemini inserted the formula in the wrong cell.	Undo, click the target cell first, then click Insert.
Gemini used XLOOKUP instead of VLOOKUP.	Both are correct. Ask 'Use VLOOKUP instead' if you want to practise VLOOKUP.
The formula works in row 2 but not lower down.	A range is not locked with $. Ask Gemini to 'make the Jobs range absolute' and fill down again.
Strong + Possible + Not suitable is not 240.	Some rows have no formula or an error. Filter column T for blanks or #N/A.

Challenge

Ask Gemini for one formula that returns the Strong count for every Position at once (hint: QUERY or a COUNTIFS against a list of positions). Check that it adds up to 88.

Reflection

Why is it better to keep the pass mark in Settings!B2 than to type 70 inside the formula?